

OLIVER BOORSTEIN

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Education

University of Oregon

Sep 2023 – Jun 2027

B.S. in Computer Science and Data Science

Eugene, Oregon

Minors: Mathematics, Music Technology

GPA: **4.21** / **4.30**

Relevant Coursework: Parallel Computing, Databases, Deep Learning, Data Structures & Algorithms, Linear Algebra

Experience

QuackHacks

May 2026 – Present

Software Engineer and Event Coordinator

Eugene, Oregon

- Led the University of Oregon's largest hackathon with over **200** hackers and a budget of **\$50,000**.
- Built production **Next.js** and **PostgreSQL** app sustaining over **10,000,000** realtime messages across **20+** tables.
- Deployed an **AWS S3** and **CloudFront** image pipeline to serve images to the QuackHacks web platform.

Oregon Networking Research Group

May 2025 – Present

Undergraduate Researcher

Eugene, Oregon

- Architected **ATLANTIS**: a **SvelteKit**, **TypeScript**, data processing and exploration platform for visualizing large-scale network telemetry, deployed on **Cloudflare Workers** with **D1** and **Drizzle ORM**.
- Built a parallelized **Python** pipeline, turning **100+ TB** NetFlow datasets to SQLite tables for **9x** faster queries.
- Ported address-analysis routines from **Haskell** to **C**, reducing processing time by **80%**.
- Developed **PyTorch GRU** pipeline to forecast networking, reducing **MAE** by **10%** over traditional methods.

Album Listening Club

Jun 2024 – Present

Co-President

University of Oregon

- Lead weekly meetings for one of Oregon's largest student clubs focused on music discussion and community.
- Manage a **\$2,000** budget to fund activities, prizes, and grow the club to over **50** weekly attendees.
- Directed local music events generating over **\$5,000** in revenue and over **1,000** attendees across events.

University of Oregon

Sep 2024 – Jun 2025

Computer Science Learning Assistant

Eugene, Oregon

- Led office hours for up to **5** students per session, covering beginner to advanced **Python** topics.
- Supported **300+** students across **three** CS courses, collaborating with TAs to improve engagement and retention.

Projects

The Stoning | *React, Convex, TypeScript, Bun*

Jan 2025 – Present

- Developed a **full-stack** music publication platform using **React**, **TypeScript**, and **Convex** in a team of 4.
- Led a site-wide migration from embedded articles to document storage in **Convex**, improving page load speeds by **20%**.
- Designed a cohesive UI with **Tailwind CSS**, improving usability and visual consistency.
- Deployed at thestoning.net via **Cloudflare** for global access and continuous integration/development.

L-Store Database | *Python, Pytest*

Sep 2025 – Dec 2025

- Directed a team of 4 to implement a lineage-based storage engine in **Python** (**80%+** test coverage via **pytest**).
- Implemented lineage-based table architecture supporting efficient OLAP/OLTP queries and contention-free merging.
- Designed concurrent transaction system using strict 2-phase locking, preventing deadlocks and ensuring thread-safety.

Awards

Vice President for Research and Innovation Fellowship

Jun 2026 – Sep 2026

University of Oregon

- Fellowship supporting my interdisciplinary research in Computer Science and Music Technology through summer 2026.

Bowerman Scholarship

Jun 2026

University of Oregon

- Nomination-based honor, awarded to only the very best of senior-level students at the University of Oregon.

Geoffrey Eric Wright Outstanding Junior Award

May 2026

University of Oregon Department of Computer Science

Technical Skills

Languages: Python, TypeScript, SQL, C/C++, JavaScript, HTML, CSS, Bash

Technologies: React, Svelte/SvelteKit, Next.js, Vite, Convex, SQLite, PostgreSQL, Drizzle ORM, NoSQL, Tailwind CSS

Tools: AWS, S3, CloudFront, Route 53, Cloudflare, D1, Workers, CI/CD, GitHub Actions, Git, Bun, Node.js, Linux/UNIX